Data-Driven Robust Performance Refinement of Dual-Stage Hard Disk Drive Controllers

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Objectives and Approach

Objective

- Design an **intuitive controller refinement algorithm** to optimize high-performance dual-stage hard disk drive (HDD) controllers.
- **Prove robust stability and performance** at every step of the algorithm.

Challenges

- **More systematic optimization** process versus indirect, experience-based, non-intuitive controller iterations.
- Representative metric of that assesses engineer's **true HDD control performance**.

Approach

- Utilize **black-box, zeroth-order (ZO) optimization algorithms** to push HDD performance.
- **Enforce robust stability** via a data-driven constraint derived from Cauchy's Argument Principle.

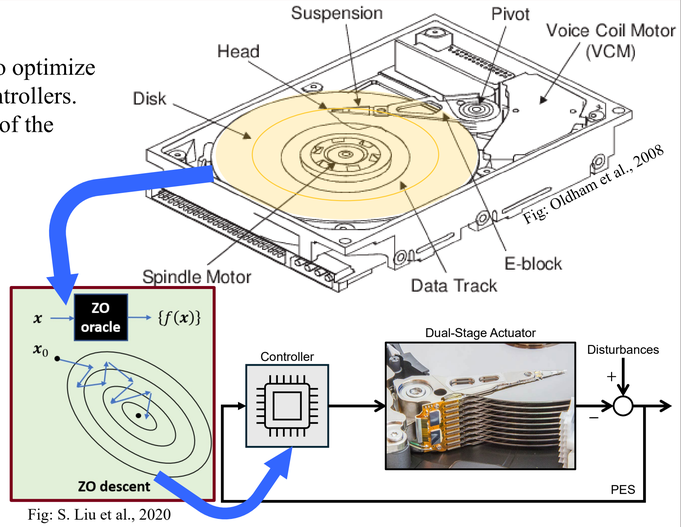
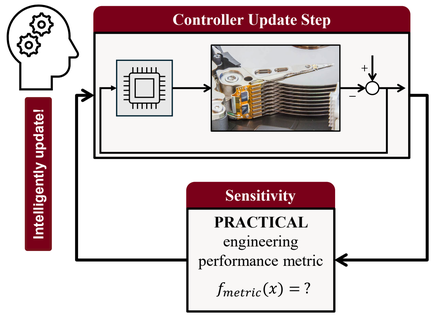


Fig: S. Liu et al., 2020

Black-Box Approach



Traditional Methods:

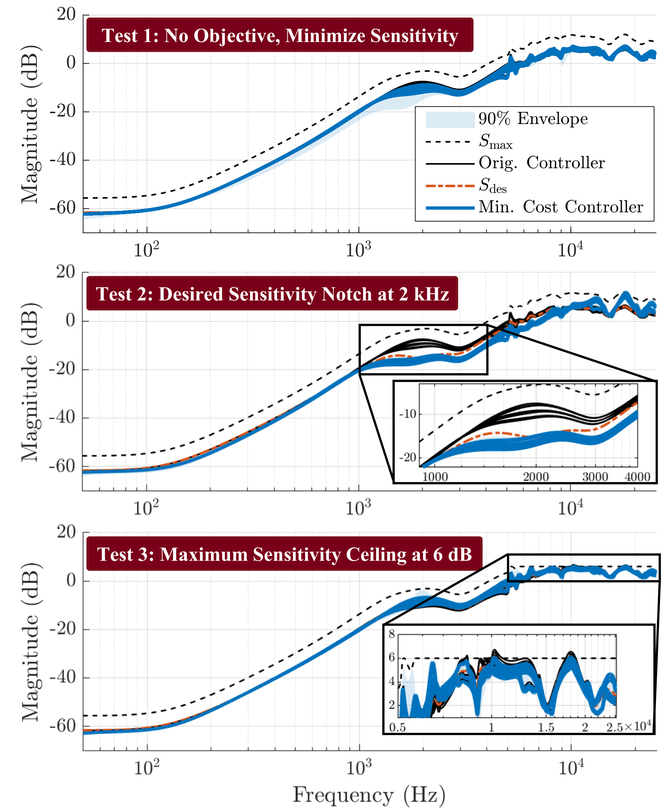
- **Conservative** robust constraints
- **Qualitative** performance assessment

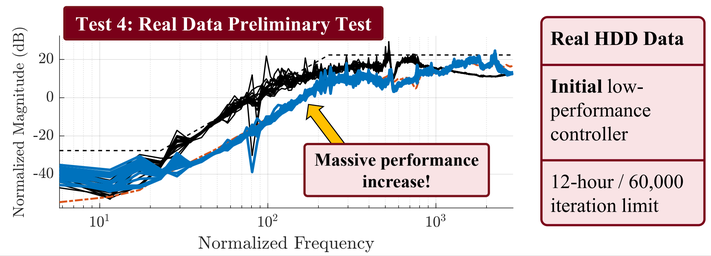
Proposed Method:

- **Direct** iteration upon controller
- **Quantitative** performance assessment
- Novel robust stability enforcement

Controller Refinement Results

Benchmark Data Set	**Initial** stable μ-syn controller	24-hour / 500,000 iter. limit





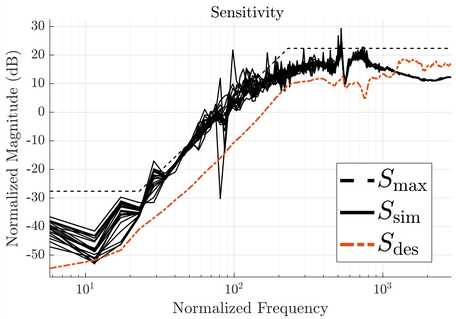
Real HDD Data

Initial low-performance controller

12-hour / 60,000 iteration limit

Zeroth-Order (ZO) Optimization and Algorithm Logic

Performance (Cost) Function Design



HDD Sensitivity:

- A control engineer's **greatest metric** of performance.

Quantitative Metrics:

- **Performance cost function** linearly scales with a desired sensitivity.
- **Maximum soft constraint** exponentially penalizes exceedingly large sensitivity responses.

Performance Cost Metric

$$\gamma_{scale} \sum_{\omega \in \Omega} \max(0, |S_{sim}|_{dB} - |S_{des}|_{dB})$$

Maximum Sensitivity Cost Metric

$$\gamma_{scale} \sum_{\omega \in \Omega} e^{\max(0,\, |S_{max}|_{dB} - |S_{sim}|_{dB})}$$

ZO Optimization Process

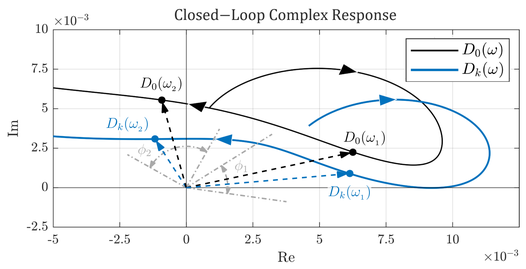
1. Initialize design vector θ_0
   $\theta_0 = [\dots, \dots, \dots, \dots] \in \mathbb{R}^n$
2. Randomize a direction vector
   $\xi \in \mathbb{R}^n \qquad \xi \sim \mathcal{N}(0, I)$
3. **Central-Point Difference**
   Take an exploration step
   $\varepsilon \in \mathbb{R}$
   $\theta_{k,hi} = \theta_k + \varepsilon\xi$
   $\theta_{k,lo} = \theta_k - \varepsilon\xi$
4. **Directional Gradient**
   Evaluate the steps
   $\nabla f(\theta_k) \approx \dfrac{f(\theta_{k,hi}) - f(\theta_{k,lo})}{2\varepsilon}$
5. Iterate on θ_k with $\alpha \in \mathbb{R}$
   $\theta_{k+1} = \theta_k - \alpha \nabla f(\theta_k)\xi$

Iterate Until Converged

Data-Driven Robust Stability!

Complex Closed-Loop Phase Constraint:

- Based on Cauchy's Argument Principle
- Enforces stability though restricting encirclements!



Proposed method benefits, from current literature:

- Data-driven **stability guarantee**
- Zeroth-order **optimization benefits**
- Includes **feasible and proper** actuator bounds

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